

Samadhan (समाधान): Solution Synopsis

MPOnline Idea & Innovation Hackathon 2026 · Problem Statement 5 · Built for all of Madhya Pradesh Repository: <https://github.com/krishnakantnagina/Samadhan> (tag `v1-mvp`) · Live URL: <https://samadhan-web.onrender.com/>

The problem

A citizen with a civic problem (no water, a broken road, uncollected garbage) has to know which department owns it, which zone or ward office to approach, and how to fill in a form or navigate a menu. Many citizens speak Hindi, Hinglish or a local dialect and would rather talk than type. A wrongly routed complaint loses days.

The solution

Samadhan is a **voice-first AI chatbot website**. The citizen speaks or types a complaint in Hindi or Hinglish. Samadhan:

1. works out what kind of message it is: a complaint, an information question, a status question, or out of context;
2. picks the right **department** (water, electricity, roads, sanitation, or the Human Evaluation desk when it is unsure);
3. asks only for what is missing (usually the issue, then the location), then asks the citizen to **confirm**;
4. files a ticket **SMD-xxxx** and routes it to the correct **ward office**, or to the district office marked `needs_review` when unsure;
5. lets the citizen check status by complaint number, typed or spoken, in the chat or on a status page.

Officers work on a **dashboard**: department-wise open counts, filters, ticket detail with the original audio, status changes, reassignment across departments with an audit trail, a review queue for low-confidence tickets, and a map.

What makes it trustworthy

The AI never has the last word. One LLM call proposes the intent, department and fields; **plain code validates every value against a service specification file** before anything is stored. The model cannot invent a department, field or office. Information questions get a fixed honest reply, a validated `.gov.in` link and a disclaimer, not a made-up answer. A citizen logs in with a mobile number before a complaint is filed, so the officer can call back and false complaints are discouraged. The public status endpoint returns only four fields.

Where it stands (measured, not estimated)

- Routing: about **90% exact** on 60 test messages (26 hand-written, 34 synthetic Bundeli/Malvi). One run each; synthetic data written by an LLM.
- **415 backend and 35 dashboard automated tests pass.**
- Voice in (Sarvam, Groq Whisper fallback), voice replies (Sarvam Bulbul), GPS, LLM fallback chain (three Groq models, then Gemini).
- Honest limits: electricity, roads, sanitation and Human Evaluation office data are **DEMO**; ward names are unverified; GPS does not yet pick a ward; mobile login uses a demo PIN, not a real SMS OTP, and there is no rate limit yet (see the roadmap).

Next

Verified ward and office data, phone-verified complaints (SMS OTP, once the SMS licence is obtained), rate limiting, then integration with MPOne's existing channels.